

PML3ZA. Regression and DL Models

UoP

DataLab, ICMAT-CSIC

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- Chapter objectives
- Regression models
- Logistic regression models: Computational problems with its Bayesian analysis
- SSMs and DLMs
- References

Chapter objectives

Objectives

- Revisit main concepts in Bayesian inference through a simple but very useful model: Linear regression
 - Prediction
- Hints on hierarchical models
- Hints on empirical Bayes
- Logistic regression. Bayesian computational problems in high dimensions
- Overview key concepts in Bayesian time series modeling with the aid of Dynamic Linear Models
- Prediction, Prediction, Prediction
- The labs reinforce concepts operationally

Regression models

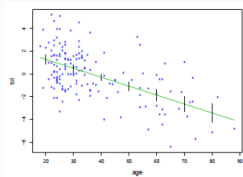
Basic setup

Problem faced: Estimating a continuous response given several explanatory variables with postulated linear relation

An example: *Consider kidney function. Data represent (x =age, y =tot, a composite measure of overall function). Kidney function declines with age and need to provide additional information concerning decline rate. Important in managing kidney transplants*

Numerical examples in Lab. Here, concepts and methods.

Check: https://en.wikipedia.org/wiki/Linear_regression



Probabilistic setup

- n observations iid with K explanatory variables, one continuous response variable.
- $\mathbf{X}$ compiles n explanatory measurements (design matrix).
- $\mathbf{y}$ compiles responses.
- Exploratory analysis suggests linear relations (possibly after transformations).
- Parameters of interest:
 - $\boldsymbol{\beta} \in \mathbb{R}^k$, regression weights,
 - σ^2 , noise variance

The postulated linear regression model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

$$\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

Classical setup

- Given $\mathbf{X}$, the likelihood of $\mathbf{y}$ is:

$$p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right).$$

- The MLEs are:

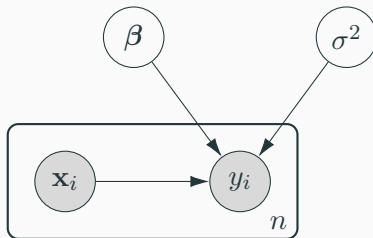
$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

$$\hat{\sigma}^2 = \frac{1}{n}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

- An unbiased estimator is:

$$\hat{\sigma}_{\text{unbiased}}^2 = \frac{1}{n - k}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

Probabilistic setup



Bayesian setup. Another source of uncertainty (or info)

- Parameters viewed as uncertain quantities
- First, use the standard non-informative (improper) prior:

$$p(\boldsymbol{\beta}, \sigma^2) \propto \frac{1}{\sigma^2}.$$

Need to aggregate both sources of information: Bayes' formula.

From prior to posterior. The joint posterior is

$$p(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}, \mathbf{X}) \propto (\sigma^2)^{-\frac{n}{2}-1} \exp \left(-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \right).$$

The posterior summarizes all information available about parameters.

The conditional marginal posteriors

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y}, \mathbf{X} \sim \mathcal{N} \left(\hat{\boldsymbol{\beta}}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \right),$$

where $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$.

$$\sigma^2 \mid \mathbf{y}, \mathbf{X} \sim \text{Inverse-Gamma} \left(\frac{n-k}{2}, \frac{1}{2} (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}})^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}) \right).$$

Sampling from the posterior

- Compute $V_\beta = (X^T X)^{-1}, \hat{\beta}, s^2$
- Sample $\sigma^2 \mid \mathbf{y}, \mathbf{X}$
- Sample $\beta \mid \sigma^2, \mathbf{y}, \mathbf{X}$

NB: Since it will be used repeatedly, it is convenient to use a QR decomposition of V_β

Posterior summarises all info available to perform

- Point estimation
- Hypothesis testing for coefficients
- Prediction
- Model checking
- Decisions

Focus on simulation

Posterior mean of β

$$\mathbb{E}[\beta \mid \mathbf{y}, \mathbf{X}] = \hat{\beta}.$$

Posterior mean of σ^2

$$\mathbb{E}[\sigma^2 \mid \mathbf{y}, \mathbf{X}] = \frac{1}{n - k - 2} (\mathbf{y} - \mathbf{X}\hat{\beta})^\top (\mathbf{y} - \mathbf{X}\hat{\beta}), \quad (n - k > 2).$$

Test for $\beta^j = 0$

- For $i = 1 \dots 1000$
 - Generate $\sigma_i^2 \sim \sigma^2 \mid \mathbf{y}, \mathbf{X}$
 - Generate $\beta_i \sim \beta \mid \sigma^2, \mathbf{y}, \mathbf{X}$
- $srt\beta^j \leftarrow sort(\beta_i^j)$
- If $0 \in [srt\beta_{50}^j, srt\beta_{950}^j]$, evidence supports $\beta^j = 0$

Basic uses. Forecasting

Given a new input $\mathbf{x}_* \in \mathbb{R}^k$

Conditioning on σ^2 , the predictive distribution is

$$y_* \mid \mathbf{x}_*, \mathbf{y}, \mathbf{X}, \sigma^2 \sim \mathcal{N} \left(\mathbf{x}_*^\top \hat{\boldsymbol{\beta}}, \sigma^2 \left(1 + \mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_* \right) \right).$$

Integrating out σ^2 , the predictive distribution becomes a Student- t :

$$y_* \mid \mathbf{x}_*, \mathbf{y}, \mathbf{X} \sim t_{n-k} \left(\mathbf{x}_*^\top \hat{\boldsymbol{\beta}}, \frac{1}{n-k} (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}})^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}) \left(1 + \mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_* \right) \right).$$

Sampling from the predictive

- Generate $\sigma_i^2 \sim \sigma^2 \mid \mathbf{y}, \mathbf{X}$
- Generate $\beta_i \sim \beta \mid \sigma^2, \mathbf{y}, \mathbf{X}$
- Generate $y_* \sim y \mid x_*, \beta, \sigma^2$

Point forecasts, Interval forecasts,...

Examining plots of residuals againsts explanatory variables, . . .
interpretable as posterior predictive checks

Advantage: Using simulation, compute the posterior predictive distribution for any data summary

Examples in Labs

Some examples of decision uses of previous models:

- Regulatory compliance: y refers to concentration of a pollutant difficult to assess; a regression model reliably predicts it from x used by an agency to impose some penalty, there being an upper bound y_0 that should not be overcome;
- candidate selection: use y to select the m highest ranked candidates; the x variables serve as predictors with y aggregating the x through a multi-objective weighted additive value model;
- resource allocation: use the whole predictive distribution to make an inflation prediction to, say, decide a budget allocation based on the maximum predictive expected utility principle.

Prior and posterior with informative conjugate prior

Prior (Normal–Inverse-Gamma)

$$\begin{aligned}\boldsymbol{\beta} \mid \sigma^2 &\sim \mathcal{N}(\boldsymbol{\beta}_0, \sigma^2 \mathbf{V}_0), \\ \sigma^2 &\sim \text{Inverse-Gamma}(a_0, b_0).\end{aligned}$$

Posterior (also Normal–Inverse-Gamma)

$$\begin{aligned}\boldsymbol{\beta} \mid \sigma^2, \mathbf{y} &\sim \mathcal{N}(\boldsymbol{\beta}_n, \sigma^2 \mathbf{V}_n), \\ \sigma^2 \mid \mathbf{y} &\sim \text{Inverse-Gamma}(a_n, b_n),\end{aligned}$$

with updated parameters

$$\begin{aligned}\mathbf{V}_n &= (\mathbf{V}_0^{-1} + \mathbf{X}^\top \mathbf{X})^{-1}, \\ \boldsymbol{\beta}_n &= \mathbf{V}_n(\mathbf{V}_0^{-1} \boldsymbol{\beta}_0 + \mathbf{X}^\top \mathbf{y}), \\ a_n &= a_0 + \frac{n}{2}, b_n = b_0 + \frac{1}{2} \left(\mathbf{y}^\top \mathbf{y} + \boldsymbol{\beta}_0^\top \mathbf{V}_0^{-1} \boldsymbol{\beta}_0 - \boldsymbol{\beta}_n^\top \mathbf{V}_n^{-1} \boldsymbol{\beta}_n \right).\end{aligned}$$

For a new input $\mathbf{x}_*$:

$$y_* \mid \mathbf{x}_*, \mathbf{y} \sim t_{2a_n} \left(\mathbf{x}_*^\top \boldsymbol{\beta}_n, \frac{b_n}{a_n} \left(1 + \mathbf{x}_*^\top \mathbf{V}_n \mathbf{x}_* \right) \right).$$

Hierarchical models. Basic setup

Before the β 's were shared. Now each observation has its β coming from a same distribution

$$\begin{aligned}\mathbf{y} \mid \beta, \sigma^2 &\sim \mathcal{N}(\mathbf{X}\beta, \sigma^2 \mathbf{I}_n), \\ \beta \mid \sigma^2, \tau^2 &\sim \mathcal{N}(\mathbf{0}, \sigma^2 \tau^2 \mathbf{I}_k).\end{aligned}$$

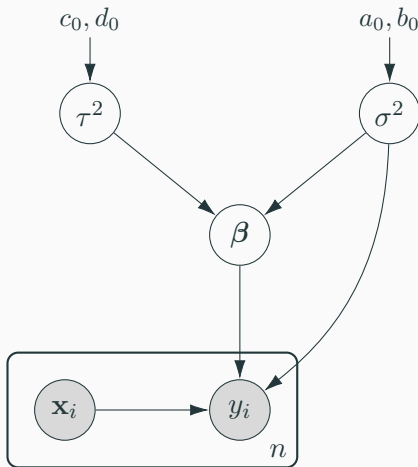
With hyperpriors

$$\begin{aligned}\sigma^2 &\sim \text{Inverse-Gamma}(a_0, b_0), \\ \tau^2 &\sim \text{Inverse-Gamma}(c_0, d_0).\end{aligned}$$

The joint posterior is

$$p(\beta, \sigma^2, \tau^2 \mid \mathbf{y}) \propto p(\mathbf{y} \mid \beta, \sigma^2) p(\beta \mid \sigma^2, \tau^2) p(\sigma^2) p(\tau^2).$$

Basic setup



- This hierarchical model induces shrinkage on β
- This is just a simple hierarchical model
- Closed-form joint posterior is not available, but conditionals are available enabling MCMC analysis.

In the previous hierarchical model, Empirical Bayes replaces the hyperprior on τ^2 by a *point estimate*, obtained from the data typically by maximizing the marginal likelihood:

$$p(\mathbf{y} \mid \tau^2, \sigma^2) = \int p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2) p(\boldsymbol{\beta} \mid \sigma^2, \tau^2) d\boldsymbol{\beta}.$$

yielding $\mathbf{y} \mid \tau^2, \sigma^2 \sim \mathcal{N}(\mathbf{0}, \sigma^2(\mathbf{I}_n + \tau^2 \mathbf{X}\mathbf{X}^\top))$, and $\hat{\tau}^2 = \arg \max_{\tau^2} p(\mathbf{y} \mid \tau^2, \sigma^2)$. Plugging $\hat{\tau}^2$ into the prior,

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y}, \hat{\tau}^2 \sim \mathcal{N}(\boldsymbol{\beta}_{EB}, \sigma^2 \mathbf{V}_{EB}),$$

where $\mathbf{V}_{EB} = (\mathbf{X}^\top \mathbf{X} + \hat{\tau}^{-2} \mathbf{I}_k)^{-1}$, $\boldsymbol{\beta}_{EB} = \mathbf{V}_{EB} \mathbf{X}^\top \mathbf{y}$

- Empirical Bayes estimates hyperparameters (here τ^2) from the data.
- The resulting estimator β_{EB} is a shrinkage estimator.
- Unlike full Bayesian inference, uncertainty in τ^2 is not propagated.
- If σ^2 is also unknown, one may estimate (τ^2, σ^2) jointly.

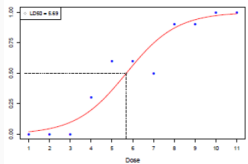
- Again we have used conjugate priors. But in several cases we were not able to derive the posteriors.... We see later on how to deal with them computationally
- Important machine learning models like LASSO or ridge regression may be seen as MAPs in Bayesian LR with certain priors.
- Linear regression models constitute the simplest example of supervised learning models for regression.
- Many NN models are non-linear regression models.

**Logistic regression models:
Computational problems with its
Bayesian analysis**

Motivation

Problem: Inference and prediction with logistic regression problems as example of GLMs

Typical example: *New anti-cancer drug being developed. Before human testing can begin, animal studies are needed to determine safe dosages. A dose-response experiment is carried out: 11 groups of 10 mice are treated with an increasing dose of drug and proportion of deaths are observed.*



Check https://en.wikipedia.org/wiki/Logistic_regression

Model formulation

Consider k dose levels $x_1, \dots, x_k$. At dose x_i , observe n_i individuals with y_i successes.

Assume

$$Y_i \sim \text{Binomial}(n_i, p_i), \quad i = 1, \dots, k$$

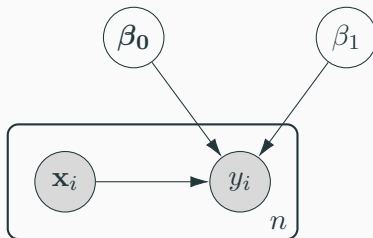
with logistic link:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 x_i$$

Equivalently:

$$p_i = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 x_i))}$$

Probabilistic setup



The likelihood function is:

$$L(\beta_0, \beta_1) = \prod_{i=1}^k \binom{n_i}{y_i} p_i^{y_i} (1 - p_i)^{n_i - y_i}$$

Substituting p_i :

$$L(\beta_0, \beta_1) = \prod_{i=1}^k \binom{n_i}{y_i} \left(\frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right)^{y_i} \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \right)^{n_i - y_i}$$

Log-likelihood:

$$\ell(\beta_0, \beta_1) = \sum_{i=1}^k \left[y_i(\beta_0 + \beta_1 x_i) - n_i \log(1 + e^{\beta_0 + \beta_1 x_i}) \right] + \text{const}$$

MLE $(\hat{\beta}_0, \hat{\beta}_1)$ satisfies:

$$\frac{\partial \ell}{\partial \beta_0} = \sum_{i=1}^k (y_i - n_i p_i) = 0$$

$$\frac{\partial \ell}{\partial \beta_1} = \sum_{i=1}^k x_i (y_i - n_i p_i) = 0$$

Nonlinear. Solve numerically .

Assume prior $\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_0 \\ \mu_1 \end{pmatrix}, \Sigma \right)$

Common choice is $\beta_0, \beta_1 \sim \mathcal{N}(0, \sigma^2)$ independent

Posterior is proportional to $p(\beta_0, \beta_1 \mid \mathbf{y}) \propto L(\beta_0, \beta_1) \cdot p(\beta_0, \beta_1)$

Thus

$$p(\beta_0, \beta_1 \mid \mathbf{y}) \propto \exp \left\{ \sum_{i=1}^k \left[y_i(\beta_0 + \beta_1 x_i) - n_i \log(1 + e^{\beta_0 + \beta_1 x_i}) \right] \right\} \times \\ \exp \left\{ -\frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\mu})^T \Sigma^{-1}(\boldsymbol{\beta} - \boldsymbol{\mu}) \right\}$$

The posterior does not have a closed form!!!

Sampling from a piecewise constant approximation

Sampling from the joint posterior through a piecewise constant approximation

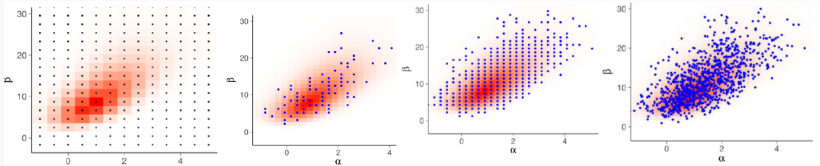
- Obtain a finite region where posterior is concentrated.
- Choose grid over region
- Evaluate density over grid
- (Normalised) Density $\times$ Cell area approximates cell prob at each cell

Sample according to grid cell probabilities

Jitter for visualisation

Sampling from a piecewise constant approximation

Sampling from the joint posterior approximately



Sampling from a piecewise constant approximation

Posteriors of derived quantities.

The case of LD50, dose level at which death probability is 0.5

$$E\left(\frac{y}{n}\right) = \theta = \text{logit}^{-1}(\beta_0 + \beta_1 x) = 0.5$$

$$\beta_0 + \beta_1 x = \text{logit}(0.5) = 0$$

$$x_{LD50} = -\frac{\beta_0}{\beta_1}$$

Sample $(\beta_0^i, \beta_1^i) \sim p(\beta_0, \beta_1 | y, n, x)$

Approximate $E(x_{LD50} | y, n, x) \approx \frac{1}{n} \sum_i^n \frac{\beta_0^i}{\beta_1^i}$

Assume parameter space is discrete and there is a true underlying distribution.

For the given parametric model, consider the parameter minimising Kullback-Leibler divergence to the true distribution.

If minimiser has positive prior probability, the posterior concentrates over the minimiser as data accumulates.

If parameter space is compact and A is a neighborhood of minimiser with positive prior probability, the posterior concentrates in A as data accumulates.

Under certain regularity conditions, the posterior approaches normality around the posterior mode as data accumulates

- Parameter space in R^p
- Mode unique
- Mode interior to parameter space
- Density continuous and positive at mode
- Log-likelihood twice differentiable at neighbourhood of mode

Asymptotics: normality with MLE

- As sample size increases, likelihood dominates (a non-degenerate) prior
- Prior behaves as approximately constant
- Posterior mode approximated by MLE
- Inverse Fisher's info matrix at MLE approximates variance

Log-likelihood $\ell(\theta) = \log L(\theta)$ and $\hat{\theta}$ the MLE

Fisher information matrix at MLE $\mathcal{I}(\hat{\theta}) = -\mathbb{E} \left[\frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta^\top} \bigg|_{\theta=\hat{\theta}} \right]$.

Inverse Fisher info approximates the covariance matrix of MLE

$$\text{Var}(\hat{\theta}) \approx \mathcal{I}(\hat{\theta})^{-1}$$

Observed Fisher info $\mathcal{J}(\hat{\theta}) = -\frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta^\top} \bigg|_{\theta=\hat{\theta}}$.

Approximate posterior by normal
Moments and regions as in normal

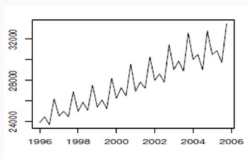
- Non-availability of posterior is the norm
- Normal asymptotics is too demanding for many cases, thus requiring other strategies
- Logistic regression as a model for classification (supervised learning with discrete response)
- A lab illustrates computationally issues

SSMs and DLMs

Problem setup

Problem faced: Forecasting the values of a time series of interest

An example: *We have available quarterly consumptions of gas in Spain. We need to forecast consumption over the next year to plan production.*



Numerical example in Lab. Here, concepts and methods.

Check summary in

<https://www.math.unm.edu/~ghuerta/tseries/dlmch2.pdf>

Three 'oldies'

- Models in time domain: ARIMA. Box Jenkins
- Models in frequency domain. Spectral analysis.
- State space models. Series as output of dynamic systems subject to random perturbations.

The Deep Learning tsunami!!!

- Recurrent NN (They are nonlinear AR!!!)
- LSTM-GRU (They are nonlinear AR!!!)
- Transformers and GPTs (They are nonlinear AR!!!)
- AutoResearch

Advantages of state space models

Natural interpretation through combination of components: trend, seasonal, regression, autoregression

ARMA embedded.

Powerful probabilistic structure facilitating flexible modeling in numerous domains

Recursive computation

Bayesian treatment naturally

Uni and multivariate

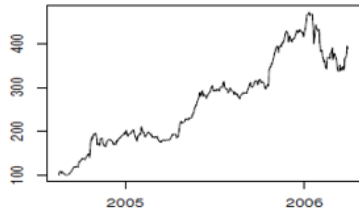
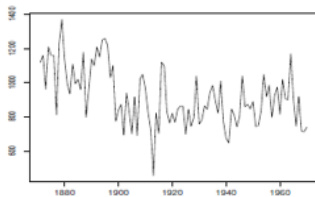
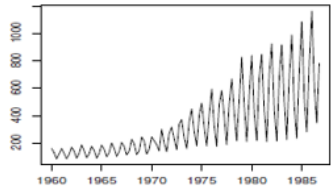
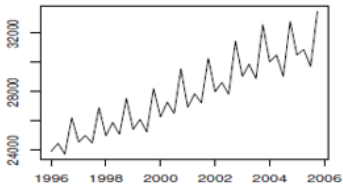
Non-stationarity, structural changes, irregular patterns naturally

Linear (Kalman filter) and non-linear (MCMC).

Very useful in applications

Explainable

Time series features



Object position from measurements subject to random errors

$$\theta \quad (Y_t : t = 1, 2, \dots)$$

$$Y_t = \theta + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

$$Y_1, Y_2, \dots | \theta \stackrel{iid}{\sim} \mathcal{N}(\theta, \sigma^2).$$

$$\theta \sim \mathcal{N}(m_0, C_0).$$

$$\theta | y_{1:n} \sim \mathcal{N}(m_n, C_n),$$

$$m_n = \mathbb{E}(\theta | y_{1:n}) = \frac{C_0}{C_0 + \sigma^2/n} \bar{y} + \frac{\sigma^2/n}{C_0 + \sigma^2/n} m_0$$

$$C_n = \text{Var}(\theta | y_{1:n}) = \left(\frac{n}{\sigma^2} + \frac{1}{C_0} \right)^{-1} = \frac{\sigma^2 C_0}{\sigma^2 + n C_0}.$$

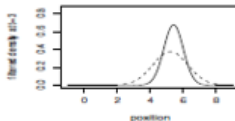
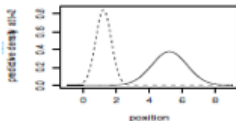
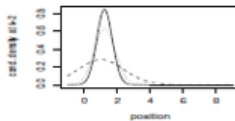
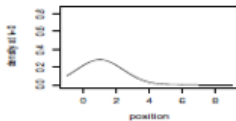
Recursively

$$\begin{aligned} m_n &= \frac{C_{n-1}}{C_{n-1} + \sigma^2} y_n + \left(1 - \frac{C_{n-1}}{C_{n-1} + \sigma^2} \right) m_{n-1} \\ &= m_{n-1} + \frac{C_{n-1}}{C_{n-1} + \sigma^2} (y_n - m_{n-1}) \end{aligned}$$

$$C_n = \left(\frac{1}{\sigma^2} + \frac{1}{C_{n-1}} \right)^{-1} = \frac{\sigma^2 C_{n-1}}{\sigma^2 + C_{n-1}}.$$

SLAM Example

Time	<u>Observation</u>	Mean	<u>Variance</u>
0	-	1	2
1	1.3	1.24	0.4
2	1.2	1.222	0.222
3			



- Introduce dynamic component. From $t=2$, object adopts a velocity

$$\theta_t = \theta_{t-1} + \nu + w_t, \quad w_t \sim \mathcal{N}(0, \sigma_w^2) \quad Y_t = \theta_t + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

With independent noise processes.

- Initial step:

$$\theta_2|y_{1:2} \sim \mathcal{N}(m_2 = 1.222, C_2 = 0.222).$$

- Prediction:

$$\theta_3|y_{1:2} \sim \mathcal{N}(a_3, R_3)$$

$$Y_3|y_{1:2} \sim \mathcal{N}(f_3, Q_3).$$

$$a_3 = E(\theta_2 + \nu + w_3|y_{1:2}) = m_2 + \nu = 5.722$$

$$f_3 = E(\theta_3 + \epsilon_3|y_{1:2}) = a_3 = 5.722$$

$$R_3 = \text{Var}(\theta_2 + \nu + w_3|y_{1:2}) = C_2 + \sigma_w^2 = 1.122$$

$$Q_3 = \text{Var}(\theta_3 + \epsilon_3|y_{1:2}) = R_3 + \sigma^2 = 1.622$$

- Initial step:

$$\theta_2|y_{1:2} \sim \mathcal{N}(m_2 = 1.222, C_2 = 0.222).$$

- Forecast:

$$\theta_3|y_{1:2} \sim \mathcal{N}(a_3, R_3)$$

$$Y_3|y_{1:2} \sim \mathcal{N}(f_3, Q_3).$$

$$a_3 = E(\theta_2 + \nu + w_3|y_{1:2}) = m_2 + \nu = 5.722$$

$$f_3 = E(\theta_3 + \epsilon_3|y_{1:2}) = a_3 = 5$$

$$R_3 = \text{Var}(\theta_2 + \nu + w_3|y_{1:2}) = C_2 + \sigma_w^2 = 1.122$$

$$Q_3 = \text{Var}(\theta_3 + \epsilon_3|y_{1:2}) = R_3 + \sigma^2$$

- Filter:

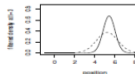
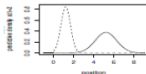
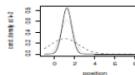
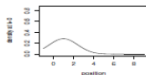
$$Y_3 = 5$$

$$e_t = y_t - f_t = -0.722$$

$$\theta_3|y_1, y_2, y_3 \sim \mathcal{N}(m_3, C_3),$$

$$m_3 = a_3 + \frac{R_3}{R_3 + \sigma^2}(y_3 - f_3) = 5.568$$

$$C_3 = \frac{\sigma^2 R_3}{\sigma^2 + R_3} = R_3 - \frac{R_3}{R_3 + \sigma^2} R_3 = 0.346$$



Observable process determined by latent process except for normal error

Latent process depends on previous state linearly, except for normal error

Forecast and estimation sequentially as data arrives

Linearity and normality define dynamic linear models (DLMs)

Time dependence structure, define state space models (SSM)

- p-variate and m-variate time series (θ_t) Y_t such that
 - (θ_t) is Markov chain
 - Given (θ_t) , Y_t are independent and just depends on θ_t

$$\begin{array}{ccccccc} \theta_0 & \rightarrow & \theta_1 & \rightarrow & \theta_2 & \rightarrow & \dots \rightarrow \theta_{t-1} & \rightarrow & \theta_t & \rightarrow & \theta_{t+1} & \rightarrow & \dots \\ & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & \downarrow \\ & & Y_1 & & Y_2 & & & & Y_{t-1} & & Y_t & & Y_{t+1} \end{array}$$

$$\pi(\theta_{0:t}, y_{1:t}) = \pi(\theta_0) \cdot \prod_{j=1}^t \pi(\theta_j | \theta_{j-1}) \pi(y_j | \theta_j)$$

- State space model with

- Observation equation

$$Y_t = F_t \theta_t + v_t, \quad v_t \sim \mathcal{N}_m(0, V_t)$$

- State equation

$$\theta_t = G_t \theta_{t-1} + w_t, \quad w_t \sim \mathcal{N}_p(0, W_t)$$

- Prior

$$\theta_0 \sim \mathcal{N}_p(m_0, C_0)$$

- State space with

$$\begin{aligned} Y_t &= \mu_t + v_t, & v_t &\sim \mathcal{N}(0, V) \\ \mu_t &= \mu_{t-1} + w_t, & w_t &\sim \mathcal{N}(0, W) \end{aligned}$$

DLM definition. Linear growth model

- State space with

$$\begin{aligned} Y_t &= \mu_t + v_t, & v_t &\sim \mathcal{N}(0, V), \\ \mu_t &= \mu_{t-1} + \beta_{t-1} + w_{t,1}, & w_{t,1} &\sim \mathcal{N}(0, \sigma_\mu^2) \\ \beta_t &= \beta_{t-1} + w_{t,2}, & w_{t,2} &\sim \mathcal{N}(0, \sigma_\beta^2) \end{aligned}$$

Inference and prediction: Key problems

Target

$$\pi(\theta_s | y_{1:t})$$

Filtering

$$s = t$$

Forecasting

$$s > t$$

Smooth

$$s < t$$

$$\begin{array}{ccccccccccc} \theta_0 & \longrightarrow & \theta_1 & \longrightarrow & \theta_2 & \longrightarrow & \cdots & \longrightarrow & \theta_{t-1} & \longrightarrow & \theta_t & \longrightarrow & \theta_{t+1} & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & \downarrow & & \\ & & Y_1 & & Y_2 & & & & Y_{t-1} & & Y_t & & Y_{t+1} & & \end{array}$$

- Predictive density for state one step ahead

$$\pi(\theta_t|y_{1:t-1}) = \int \pi(\theta_t|\theta_{t-1})\pi(\theta_{t-1}|y_{1:t-1}) d\theta_{t-1}$$

- Predictive density

$$\pi(y_t|y_{1:t-1}) = \int \pi(y_t|\theta_t)\pi(\theta_t|y_{1:t-1}) d\theta_t$$

- Filtering density

$$\pi(\theta_t|y_{1:t}) = \frac{\pi(y_t|\theta_t)\pi(\theta_t|y_{1:t-1})}{\pi(y_t|y_{1:t-1})}$$

- Currently,

$$\theta_{t-1}|y_{1:t-1} \sim \mathcal{N}(m_{t-1}, C_{t-1})$$

- State predictive density one step ahead, normal

$$a_t = \mathbb{E}(\theta_t|y_{1:t-1}) = G_t m_{t-1},$$

$$R_t = \text{Var}(\theta_t|y_{1:t-1}) = G_t C_{t-1} G_t' + W_t$$

- Predictive density, normal

$$f_t = \mathbb{E}(Y_t|y_{1:t-1}) = F_t a_t,$$

$$Q_t = \text{Var}(Y_t|y_{1:t-1}) = F_t R_t F_t' + V_t$$

- Filtering density

$$m_t = \mathbb{E}(\theta_t|y_{1:t}) = a_t + R_t F_t' Q_t^{-1} e_t,$$

$$e_t = Y_t - f_t$$

$$C_t = \text{Var}(\theta_t|y_{1:t}) = R_t - R_t F_t' Q_t^{-1} F_t R_t$$

- Looking ahead in the future

$$\begin{array}{ccccccc} \theta_t & \longrightarrow & \theta_{t+1} & \longrightarrow & \dots & \longrightarrow & \theta_{t+k} \\ | & & & & & & | \\ Y_{1:t} & & & & & & Y_{t+k} \end{array}$$

- State k steps ahead

$$\pi(\theta_{t+k}|y_{1:t}) = \int \pi(\theta_{t+k}|\theta_{t+k-1})\pi(\theta_{t+k-1}|y_{1:t}) d\theta_{t+k-1}$$

- Observation k steps ahead

$$\pi(y_{t+k}|y_{1:t}) = \int \pi(y_{t+k}|\theta_{t+k})\pi(\theta_{t+k}|y_{1:t}) d\theta_{t+k}$$

- State, normal with mean and variance

$$\begin{aligned}a_t(k) &= G_{t+k}a_{t,k-1}, \\ R_t(k) &= G_{t+k}R_{t,k-1}G'_{t+k} + W_{t+k}\end{aligned}$$

- Observation, normal with mean and variance

$$\begin{aligned}f_t(k) &= F_{t+k}a_t(k), \\ Q_t(k) &= F_{t+k}R_t(k)F'_{t+k} + V_t\end{aligned}$$

- Prediction errors

$$e_t = Y_t - E(Y_t | y_{1:t-1}) = Y_t - f_t \qquad e_t = Y_t - F_t a_t = F_t \theta_t + v_t - F_t a_t \\ = F_t(\theta_t - a_t) + v_t.$$

- Properties facilitating validation
 - Expected value is 0
 - Observations from gaussian process
 - $\tilde{e}_t = e_t / \sqrt{Q_t}$ White noise gaussian process

Modeling: Superposition principle

- The sum of independent DLMs is a DLM

$$Y_t = Y_{1,t} + \dots + Y_{h,t}$$

$$\begin{aligned} Y_{i,t} &= F_{i,t}\theta_{i,t} + v_{i,t}, & v_{i,t} &\sim \mathcal{N}(0, V_{i,t}), \\ \theta_{i,t} &= G_{i,t}\theta_{i,t-1} + w_{i,t}, & w_{i,t} &\sim \mathcal{N}(0, W_{i,t}) \end{aligned}$$

results in

$$\begin{aligned} Y_t &= F_t\theta_t + v_t, & v_t &\sim \mathcal{N}(0, V_t), \\ \theta_t &= G_t\theta_{t-1} + w_t, & w_t &\sim \mathcal{N}(0, W_t), \end{aligned}$$

$$\theta_t = \begin{bmatrix} \theta_{1,t} \\ \vdots \\ \theta_{h,t} \end{bmatrix} \quad F_t = [F_{1,t} | \dots | F_{h,t}] \quad G_t = \begin{bmatrix} G_{1,t} & & \\ & \ddots & \\ & & G_{h,t} \end{bmatrix} \quad W_t = \begin{bmatrix} W_{1,t} & & \\ & \ddots & \\ & & W_{h,t} \end{bmatrix} \quad V_t = \sum_{i=1}^j V_{i,t}$$

Model building strategy based on blocks

- Order n

$$\begin{cases} Y_t = \theta_{t,1} + v_t \\ \theta_{t,j} = \theta_{t-1,j} + \theta_{t-1,j+1} + w_{t,j} & j = 1, \dots, n-1 \\ \theta_{t,n} = \theta_{t-1,n} + w_{t,n} \end{cases}$$

$$F = (1, 0, \dots, 0)$$
$$G = \begin{bmatrix} 1 & 1 & 0 & \dots & 0 \\ 0 & 1 & 1 & 0 & \dots & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & \dots & 0 & 1 & 1 \\ 0 & \dots & & 0 & 1 \end{bmatrix}$$
$$W = \text{diag}(W_1, \dots, W_n).$$

- Predictive function

$$f_t(k) = E(Y_{t+k} | y_{1:t}) = a_{t,0} + a_{t,1}k + \dots + a_{t,n-1}k^{n-1}$$

- Consider cases $n=1,2,3$

- State vector dimension $s-1$

$$F = (1, 0, \dots, 0)$$

$$G = \begin{bmatrix} -1 & -1 & \dots & -1 & -1 \\ 1 & 0 & & 0 & 0 \\ 0 & 1 & & 0 & 0 \\ & & \ddots & & \\ 0 & 0 & & 1 & 0 \end{bmatrix}.$$

NB: Better results with Fourier decomposition (retain only meaningful components)

- From the standard regression model

$$Y_t = \beta_1 + \beta_2 x_t + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

- To the dynamic regression model

$$Y_t = \beta_{1,t} + \beta_{2,t} x_t + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

$$\begin{aligned} Y_t &= x_t' \theta_t + v_t, & v_t &\sim \mathcal{N}(0, \sigma_t^2) \\ \theta_t &= G_t \theta_{t-1} + w_t, & w_t &\sim \mathcal{N}_p(0, W_t) \end{aligned}$$

- Trend + Seasonal + Regression term+ Low order AR term (1)

θ_t

`dlmModPoly(2) + dlmModSeas(4)`

- Again we have gone through the 'easy' models.
- U or V unknown?
- Non-normal?
- Non-linear? e.g. NNs: Transformers, GPTs,...

References

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PML Reading group Check Session 5

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Lab 3.1. Linear Regression

Lab 3.2 Numerical strategies with logistic regression

Lab 3.3 DLMS in R with dlm